

1

A stock price is currently \$40. It is known that at the end of one month it will be either \$42 or \$38. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a one-month European call option with a strike price of \$39?

Following the way that we set up the portfolios in the text, we consider a portfolio with Δ shares of a stock and short 1 call option with the specification as in the problem. The portfolio value would be the following in each situation:

- If the price becomes \$42: $42\Delta - 3$
- If the price becomes \$38: 38Δ

As the portfolio is riskless when these are equal, we have

$$42\Delta - 3 = 38\Delta$$

$$4\Delta = 3$$

$$\Delta = 0.75$$

So the riskless portfolio is longing 0.75 shares and shorting 1 call option. The current value of the portfolio is therefore $38 \cdot 0.75 = 28.5$, so with discounting at the risk-free rate over a month, we have

$$(0.75 \cdot 40 - f)e^{0.08 \cdot \frac{1}{12}} = 28.5$$

$$(30 - f)e^{\frac{1}{150}} = 28.5$$

$$30 - f = 28.3106319284$$

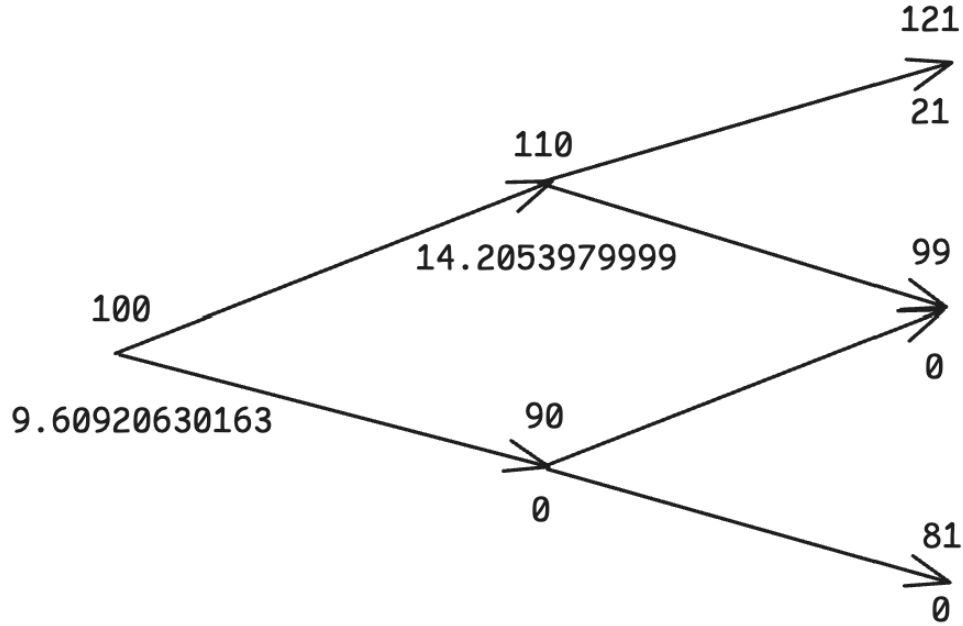
$$f = 1.6893680716 \approx \boxed{\$1.69}$$

2

A stock price is currently \$100. Over each of the next two six-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 8% per

annum with continuous compounding. What is the value of a one-year European call option with a strike price of \$100?

In this case, we have $u = 1.10, d = 0.90, \Delta t = 0.5, r = 0.08$. We set up the tree like so:



We know that

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

$$p = \frac{e^{0.08 \cdot 0.5} - 0.9}{1.10 - 0.90}$$

$$p = \frac{e^{0.04} - 0.9}{0.2}$$

$$p = \frac{0.14081077419}{0.2} = 0.70405387095$$

We use this to work backwards in the tree. First step:

$$f_u = e^{-0.08 \cdot 0.5} \cdot ((0.70405387095 \cdot 21) + ((1 - 0.70405387095) \cdot 0))$$

$$= e^{-0.04} \cdot 14.78513129 = 14.2053979999$$

And last step to the beginning:

$$f = e^{-0.08 \cdot 0.5} \cdot ((0.70405387095 \cdot 14.2053979999) + ((1 - 0.70405387095) \cdot 0))$$

$$= e^{-0.04} \cdot 10.0013654502 = 9.60920630163 \approx \boxed{\$9.61}$$

3

For the situation considered in Problem 2, what is the value of a one-year European put option with a strike price of \$100? Verify that the European call and European put prices satisfy put-call parity.

With the same setup as above, we can recreate the tree or just calculate the option value directly:

$$\begin{aligned} f &= e^{-r\Delta t}[pf_u + (1-p)f_d] \\ &= e^{-2r\Delta t}[p^2f_{uu} + 2p(1-p)f_{ud} + (1-p)^2f_{dd}] \\ &= e^{-2 \cdot 0.08 \cdot 0.5}[0.70405387095 \cdot 0 + 2 \cdot 0.70405387095 \cdot (1 - 0.70405387095) \cdot 1 + (1 - 0.70405387095)^2 \cdot 19] \\ &= e^{-0.08}[2 \cdot 0.70405387095 \cdot 0.29594612905 + 0.29594612905^2 \cdot 19] \\ &= 0.92311634638 \cdot (0.4167240355 + 1.66409811451) \\ &= 1.92084094058 \approx \boxed{\$1.92} \end{aligned}$$

Put-call parity tells us that

$$\begin{aligned} c + Ke^{-rT} &= p + S_0 \\ 9.61 + 100e^{-0.08 \cdot 1} &= 1.92 + 100 \\ 9.61 + 92.311634638 &= 101.92 \\ 101.921634638 &\approx 101.92 = 101.92 \end{aligned}$$

which indeed shows us that put-call parity does hold!

4

A stock price is currently \$40. It is known that at the end of three months it will be either \$45 or \$35. The risk-free rate of interest with quarterly compounding is 8%

per annum. Calculate the value of a three-month European put option on the stock with an exercise price of \$40. Verify that no-arbitrage arguments and risk-neutral valuation arguments give the same answers.

We set up a portfolio with Δ shares and 1 option, which works in this case since we are working with a put option, which is essentially negative. So we know that the value of the portfolio would be the following in each situation:

- If the price becomes \$45: 45Δ
- If the price becomes \$35: $35\Delta - 5$

As the portfolio is riskless when these are equal, we have

$$45\Delta = 35\Delta - 5$$

$$10\Delta = -5$$

$$\Delta = -0.5$$

So the riskless portfolio is essentially short 0.5 shares and longing 1 put option. We see that the value of the portfolio is $45 \cdot -0.5 = -22.5$. Discounting at the risk-free rate over the 3 months, we can solve for f accordingly:

$$(-0.5 \cdot 40 + f) \cdot (1 + 0.08 \cdot \frac{3}{12}) = -22.5$$

$$(-20 + f) \cdot 1.02 = -22.5$$

$$-20 + f = -22.0588235294$$

$$f = 2.0588235294 \approx \boxed{\$2.06}$$

We confirm this using the risk-neutral valuation argument:

$$45p + 35(1 - p) = 40 \cdot (1 + 0.08 \cdot \frac{3}{12})$$

$$45p + 35 - 35p = 40 \cdot 1.02$$

$$10p = 40.8 - 35$$

$$p = 0.58$$

So the expected value of the option is

$$\begin{aligned} & 0.58 \cdot 0 + (1 - 0.58) \cdot 5 \\ & = 0 + 0.42 \cdot 5 = 2.10 \end{aligned}$$

and adjusting for present value is $2.10 \div 1.02 = 2.05882352941 \approx 2.06$, which is consistent with what we previously found using the no-arbitrage argument.

5

Consider a European call option on a non-dividend-paying stock where the stock price is \$40, the strike price is \$40, the risk-free rate is 4% per annum, the volatility is 30% per annum, and the time to maturity is six months.

(a)

Calculate u , d , and p for a two step tree

We do a two step tree over six months, so we have $\Delta t = 0.5 \div 2 = 0.25$. We therefore have

$$u = e^{0.3 \cdot \sqrt{0.25}} = e^{0.15} = 1.16183424273$$

and

$$d = \frac{1}{u} = \frac{1}{1.16183424273} = 0.86070797642$$

Finally:

$$\begin{aligned} p &= \frac{e^{r\Delta t} - d}{u - d} \\ &= \frac{e^{0.04 \cdot 0.25} - 0.86070797642}{1.16183424273 - 0.86070797642} \\ p &= \frac{e^{0.01} - 0.86070797642}{0.30112626631} = \frac{0.14934219066}{0.30112626631} = 0.49594541349 \end{aligned}$$

(b)

Value the option using a two step tree. We fill in the tree using the values we found in part *a*, with the prices on the tree as follows:

- $S_u = 40u = 40 \cdot 1.16183424273 = 46.4733697092$
- $S_{uu} = 40u^2 = 40 \cdot 1.16183424273^2 = 53.9943523032$
- $S_{ud} = S_{du} = 40du = 40 \cdot 1.16183424273 \cdot 0.86070797642 = 39.9999999998 \approx 40$
- $S_d = 40d = 40 \cdot 0.86070797642 = 34.4283190568$
- $S_{dd} = 40d^2 = 40 \cdot 0.86070797642^2 = 29.6327288268$

And the corresponding f values:

$$\begin{aligned} f_u &= e^{-0.04 \cdot \frac{1}{2} \div 2} \cdot [(0.49594541349 \cdot 13.9943523032) + (1 - 0.49594541349) \cdot 0] \\ &= e^{-0.01} \cdot (0.49594541349 \cdot 13.9943523032) \\ f_u &= 0.99004983374 \cdot 6.94043483954 = 6.87137635897 \end{aligned}$$

and

$$\begin{aligned} f &= e^{-0.04 \cdot \frac{1}{2} \div 2} \cdot [(0.49594541349 \cdot 6.87137635897) + (1 - 0.49594541349) \cdot 0] \\ f &= 0.99004983374 \cdot 3.40782758959 = 3.37391913849 \approx \boxed{\$3.37} \end{aligned}$$

(c)

Verify that DerivaGem gives the same answer.

(d)

Use DerivaGem to value the option with 5, 50, 100, and 500 time steps.